RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Wednesday, 7 October 2026, 9:05 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:06 AM
Time taken	21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int max_value(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    char s1[1001];
    char s2[1001];

    if (scanf("%s", s1) != 1) return 0;
    if (scanf("%s", s2) != 1) return 0;

    int m = strlen(s1);
    int n = strlen(s2);

    int **dp = (int **)malloc((m + 1) * sizeof(int *));
    for (int i = 0; i <= m; i++) {
        dp[i] = (int *)calloc((n + 1), sizeof(int));
    }

    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = max_value(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

    printf("%d\n", dp[m][n]);

    for (int i = 0; i <= m; i++) {
```

```
36 ▾    for (int i = 0; i <= m; i++) {
37          free(dp[i]);
38      }
39      free(dp);
40
41      return 0;
42  }
43
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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